Maths Exploring the Exercises at Terminal-07

Progressive exercises

E1 · Direct application. Two events are considered A and B such that $P(A) = 0{,}3$, $P_A(B) = 0{,}8$ and $P_{\overline{A}}(B) = 0{,}1$. Calculate $P(B)$then $P_B(A)$.

E2 · Application and justification. Random variable X follows the binomial distribution of parameters 8 and $\frac{1}{4}$. Give the exact values of $P(X = 3)$, $E(X)$ and $V(X)$.

E3 · Method to be organised. Let X a random variable with integer values. k, we're laying $F(k) = P(X \leq k)$. Express $P(X \geq 4)$ and $P(2 \leq X \leq 5)$ with the help of F.

E4 · Choice of strategy. Five cards are drawn successively, without discount, in a set of 52 cards containing four aces. X the number of obtained. The variable X Does she follow a binomial distribution? Justify.

E5 · Synthesis or demonstration. We repeat independently a test whose probability of success is $\frac{1}{2}$. Determine the smallest number of tests for which the probability of obtaining less than 0.9% success.